

# claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

## Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-  
a490bfd6fedadac89.jsonl`  
- SHA-256: `1316a0ca8a6e3d38213cd5c69c58de92ab0bec4bff6be92c3c8d4d8131f35748`  
- Source modified: `2026-06-16T03:22:56+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

## Transcript

1. user (2026-06-16T03:21:40.625Z)

### Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

### Research question

Research question: **\*\*What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?\*\*** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best **\*recent\*** methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

## Claim under review

"TriDet's Trident-head localizes action boundaries by estimating a relative probability distribution over a set of local 'bins' around the boundary (computing the offset as an expected value over neighboring instants), rather than regressing a boundary offset from a single instant or from a segment's global feature ? directly addressing imprecise/ambiguous boundary localization."

**\*\*Source:\*\*** [https://openaccess.thecvf.com/content/CVPR2023/papers/Shi\\_TriDet\\_Temporal\\_Action\\_Detection\\_With\\_Relative\\_Boundary\\_Modeling\\_CVPR\\_2023\\_paper.pdf](https://openaccess.thecvf.com/content/CVPR2023/papers/Shi_TriDet_Temporal_Action_Detection_With_Relative_Boundary_Modeling_CVPR_2023_paper.pdf) (primary)

**\*\*Supporting quote:\*\*** "instead of directly predicting the boundary offsets based on the center point feature, the proposed Trident-head models the action boundary via an estimated relative probability distribution of the boundary (see Fig. 1). The boundary offset is then computed based on the expected values of neighboring locations (i.e. bins)."

## Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

**\*\*refuted=true\*\*** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

**\*\*refuted=false\*\*** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

## 2. assistant (2026-06-16T03:21:44.491Z)

I'll verify this claim about TriDet's Trident-head boundary modeling mechanism. Let me fetch the primary source.

## 3. assistant (2026-06-16T03:21:47.662Z)

I have the tools loaded. Let me fetch the primary source to verify the claim about TriDet's Trident-head.

## 4. user (2026-06-16T03:21:57.124Z)

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

## 5. assistant (2026-06-16T03:21:59.143Z)

The CVPR open-access PDF blocked the fetch. Let me try the arXiv version and search for corroborating descriptions.

## 6. user (2026-06-16T03:22:06.281Z)

# Response Based on Available Content

Based on the provided arXiv webpage, the abstract describes TriDet's approach but with limited technical detail:

## What the Abstract States

The paper proposes a "novel Trident-head to model the action boundary via an estimated relative probability distribution around the boundary." This directly addresses question (1)?yes, it estimates a relative probability distribution around boundaries.

## What Cannot Be Confirmed

From the abstract alone, I cannot definitively confirm:

- **(2)** Whether boundary offsets are computed as expected values over neighboring locations
- **(3)** Whether direct offset regression from center points is explicitly contrasted
- **(4)** The specific problem statement beyond the general issue

## Available Problem Statement

The abstract states the core motivation: "Existing methods often suffer from imprecise boundary predictions due to the ambiguous action boundaries in videos."

The Trident-head approach is framed as alleviating this problem through probabilistic boundary modeling rather than presumably deterministic point-based regression, but the full technical details distinguishing this approach would require accessing the complete paper PDF.

## 7. user (2026-06-16T03:22:12.037Z)

Web search results for query: "TriDet Trident-head relative boundary modeling probability distribution bins expected value CVPR 2023"

Links: [{"title": "TriDet - CVPR 2023 Open Access Repository", "url": "https://openaccess.thecvf.com/content/CVPR2023/html/Shi\_TriDet\_Temporal\_Action\_Detection\_With\_Relative\_Boundary\_Modeling\_CVPR\_2023\_paper.html"}, {"title": "TriDet: Temporal Action Detection", "url": "https://www.emergentmind.com/papers/2303.07347"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling | IEEE Conference Publication | IEEE Xplore", "url": "https://ieeexplore.ieee.org/document/10203543/"}]

Temporal Action Detection With Relative Boundary Modeling", "url": "https://cvpr.thecvf.com/virtual/2023/poster/21956"}, {"title": "[2303.07347] TriDet: Temporal Action Detection with Relative Boundary Modeling", "url": "https://arxiv.org/abs/2303.07347"}, {"title": "TriDet: Temporal Action Detection with Relative Boundary Modeling Dingfeng Shi?", "url": "https://openaccess.thecvf.com/content/CVPR2023/papers/Shi\_TriDet\_Temporal\_Action\_Detection\_With\_Relative\_Boundary\_Modeling\_CVPR\_2023\_paper.pdf"}, {"title": "[Quick Review] TriDet: Temporal Action Detection with Relative Boundary Modeling", "url": "https://liner.com/review/tridet-temporal-action-detection-with-relative-boundary-modeling"}, {"title": "GitHub - dingfengshi/TriDet: [CVPR2023] Code for the paper, TriDet: Temporal Action Detection with Relative Boundary Modeling", "url": "https://github.com/dingfengshi/TriDet"}, {"title": "TriDet/README.md at master · dingfengshi/TriDet", "url": "https://github.com/dingfengshi/TriDet/blob/master/README.md"}]

Based on the search results, here's information about TriDet and the Trident-head approach from the CVPR 2023 paper:

## Overview

TriDet is a one-stage framework for temporal action detection that addresses imprecise boundary predictions by proposing a novel Trident-head to model the action boundary via an estimated relative probability distribution around the boundary.

## Trident-head Mechanism

The Trident-head consists of three components: a start head, an end head, and a center-offset head, which are designed to locate the start boundary, end boundary, and the temporal center of the action, respectively.

The key innovation involves probability distribution and expected value calculations. The predicted boundary response in neighboring bin set is summed element-wise with the center offset corresponding to the instant, which is further estimated as the relative boundary distribution, and finally, the offset is computed based on the expected value of the bin.

## Performance

TriDet hits an average mAP of 69.3% on THUMOS14, outperforming the previous best by 2.5%, but with only 74.6% of its latency.

## Additional Features

In the feature pyramid of TriDet, a Scalable-Granularity Perception (SGP) layer aggregates information across different temporal granularities, which is much more efficient than recent transformer-based feature pyramids.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

### 8. assistant (2026-06-16T03:22:15.636Z)

The arXiv abstract confirms the core: "model the action boundary via an estimated relative probability distribution around the boundary" and the motivation "imprecise boundary predictions due to the ambiguous action boundaries." The search corroborates the "bins" and "expected value of the bin" mechanism. Let me get the verbatim Trident-head detail from the paper body to confirm the contrast against single-instant/global regression and the "expected values of neighboring locations (i.e. bins)" wording the claim relies on.

### 9. user (2026-06-16T03:22:23.605Z)

I've reviewed the provided PDF content, but it appears to be a technical document (likely a research paper) with encoded/compressed streams. The content is not in plain text format, making it impossible to locate and quote the specific sentences about "Trident-head and relative boundary modeling."

To accurately provide the verbatim quotes you've requested?specifically about:

1. Boundary modeling via estimated relative probability distribution

2. Boundary offset computation from expected values
3. Contrasts with offset regression approaches
4. Bin definitions around boundaries

I would need access to the document in a readable text format. The PDF structure visible here contains mostly binary/compressed data that cannot be reliably parsed for exact quotations.

**\*\*Recommendation:\*\*** Please provide either:

- A plain text excerpt from the relevant section, or
- The document's source (paper title/link) so I can reference the published version

This ensures I can deliver accurate, properly-attributed quotes within your specified constraints.

[Binary content (application/pdf, 1.4MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579910495-xf7r2b.pdf]

#### 10. assistant (2026-06-16T03:22:25.598Z)

The PDF was saved locally. Let me read the relevant pages directly to get the verbatim text.

#### 11. user (2026-06-16T03:22:27.155Z)

pdftoppm failed: Command 'pdftoppm' not found or is in an unsafe location (current directory)

#### 12. assistant (2026-06-16T03:22:29.791Z)

PDF rendering isn't available in this environment. Let me try extracting text via Python, which is more reliable for getting verbatim quotes.

#### 13. user (2026-06-16T03:22:34.576Z)

=== Trident-head ===

for temporal action detection. Existing methods often suffer from imprecise boundary predictions due to the ambiguous action boundaries in videos. To alleviate this problem, we propose a novel Trident-head to model the action boundary via an estimated relative probability distribution around the boundary. In the feature pyramid of TriDet, we propose an efficient Scalable-Granularity Perception (SGP) la

=== relative probability ===

suffer from imprecise boundary predictions due to the ambiguous action boundaries in videos. To alleviate this problem, we propose a novel Trident-head to model the action boundary via an estimated relative probability distribution around the boundary. In the feature pyramid of TriDet, we propose an efficient Scalable-Granularity Perception (SGP) layer to mitigate the rank loss problem of self-attention that takes

=== expected value ===

in set is summed element-wise with the center offset corresponding to the instant  $t$ , which is further estimated as the relative boundary distribution. Finally, the offset is computed based on the expected value of the bin. temporal length [49] of the action at each instant of the feature and refine with the boundary feature [21, 34], or [22, 23, 48] simply predict an actionness score (indicating the probabi

=== center point feature ===

one-stage action detector with a novel detection head named Trident-head tailored for action boundary localization. Specifically, instead of directly predicting the boundary offsets based on the center point feature, the proposed Trident-head models the action boundary via an estimated relative probability distribution of the boundary (see Fig. 1). The boundary offset is then computed based on the expected value

=== bins ===

s the action boundary via an estimated relative probability distribution of the boundary (see Fig. 1). The boundary offset is then computed based on the expected values of neighboring

locations (i.e. bins). Apart from the Trident-head, in this work, the proposed action detector consists of a backbone network and a feature pyramid. Recent TAD methods [10, 42, 49] adopt the transformer-based feature p

=== neighboring ===

proposed Trident-head models the action boundary via an estimated relative probability distribution of the boundary (see Fig. 1). The boundary offset is then computed based on the expected values of neighboring locations (i.e. bins). Apart from the Trident-head, in this work, the proposed action detector consists of a backbone network and a feature pyramid. Recent TAD methods [10, 42, 49] adopt the transf

#### 14. assistant (2026-06-16T03:22:47.088Z)

I now have the verbatim text from the primary source PDF. All elements of the claim are directly confirmed.

Let me verify each component of the claim against the extracted verbatim text:

1. **"estimating a relative probability distribution over a set of local 'bins' around the boundary"** ? Confirmed: "models the action boundary via an estimated relative probability distribution of the boundary... The boundary offset is then computed based on the expected values of neighboring locations (i.e. bins)."
2. **"computing the offset as an expected value over neighboring instants"** ? Confirmed: "the offset is computed based on the expected value of the bin" and "computed based on the expected values of neighboring locations (i.e. bins)."
3. **"rather than regressing a boundary offset from a single instant"** ? Confirmed: "instead of directly predicting the boundary offsets based on the center point feature, the proposed Trident-head models the action boundary via an estimated relative probability distribution."
4. **"directly addressing imprecise/ambiguous boundary localization"** ? Confirmed: "Existing methods often suffer from imprecise boundary predictions due to the ambiguous action boundaries in videos. To alleviate this problem, we propose a novel Trident-head."

The only minor nuance: the claim adds "or from a segment's global feature" as an alternative the paper contrasts against. The paper's explicit contrast is specifically against "directly predicting the boundary offsets based on the center point feature." The "single instant" / "center point feature" contrast is verbatim-supported; the "segment's global feature" extension is a reasonable generalization (BSN/BMN-style global/RoI regression) but is not the specific contrast the paper draws. This is additive framing, not a misrepresentation of the paper's own stated mechanism. The core mechanistic claim is fully and verbatim supported by the primary source.

Adversarial check results:

- **Quote support**: Strong ? the supporting quote matches the PDF verbatim, and surrounding sentences confirm every sub-claim (bins, expected value, contrast with center-point regression, imprecise-boundary motivation).
- **Contradicting evidence**: None found. WebSearch corroborated via the CVPR repository, arXiv, and independent reviews (emergtmind, liner).
- **Source quality**: Primary source (CVPR 2023 peer-reviewed paper, official arXiv PDF). Matches the claim's strength.
- **Currency**: Current (2023), within the requested 2022-2025 window and still the relevant state of the art.
- **Marketing/cherry-pick**: No ? this is a mechanistic architectural description from the methods section, not a benchmark cherry-pick.

The claim is well-supported, current, and from a primary source. Not refuted.

#### 15. user (2026-06-16T03:22:53.741Z)

Structured output provided successfully

#### 16. assistant (2026-06-16T03:22:56.700Z)

Verified. The claim is **\*\*not refuted\*\*** ? it is fully supported by the primary source (CVPR 2023 / arXiv 2303.07347), confirmed via verbatim text extraction from the paper PDF, with the only nuance being a harmless additive generalization ("segment's global feature") that does not misrepresent the paper's actual mechanism.